

Data Mining

Dimensionality Reduction: Part 2

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Outline

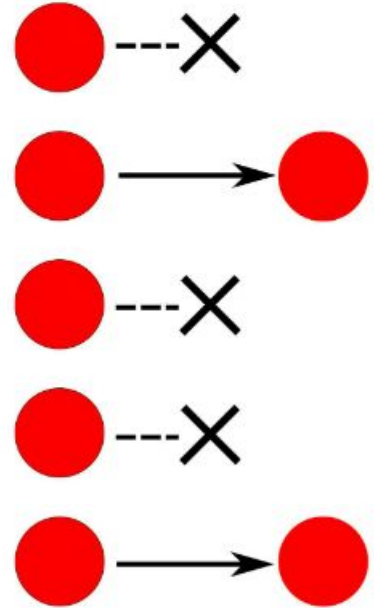
- What is Feature Subset Selection?
- Why doing Feature Subset Selection?
- Taxonomy of Feature Subset Selection methods
 - Unsupervised methods
 - Supervised methods
 - Filter methods
 - Wrapper methods
- Comparison of Feature Subset Selection methods

What is Feature Subset Selection (FSS)?

Definition: select a subset of features (attributes) from a given feature set.

Goal: optimize an objective function for the chosen features.

- **Feature extraction (e.g. PCA):**
 - Transforming the existing features into a lower dimensional space.
- **Feature selection:**
 - Selecting a subset of the existing features without transformation.



What is Feature Subset Selection (FSS)?

Feature Set:

$$X = \{x_i \mid i = 1, \dots, n\}$$

Objective:

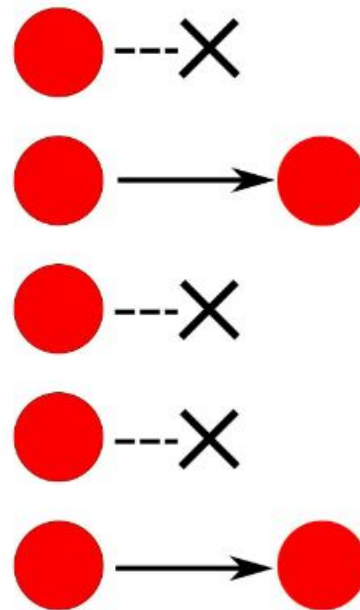
Find a subset of features:

$$Y_m = \{x_{i_1}, x_{i_2}, \dots, x_{i_m}\} : m < n$$

that optimizes the objective function $J(Y_m)$

Optimization:

$$Y = \arg \max_{m, i_m} J(Y_m)$$



Why doing Feature Subset Selection?

*Why not stick to feature extraction,
as it essentially accomplishes the task
of dimensionality reduction?*

Why doing Feature Subset Selection?

- **Feature values may be expensive to collect and obtain**
 - Test many features (sensors) and choose only a few for the final implementation.
- **The need to extract meaningful rules from data mining models**
 - When you transform the original features into new ones, the measurement units of the features are lost.
- **Features may not be numeric**
 - A typical situation in data mining.
- **Fewer features means fewer parameters for pattern recognition**
 - Improved the generalization capabilities of the model and avoid overfitting.
 - Reduced model complexity and running time.

Why Feature selection is challenging?

- Number of features = 3
 - {A1, A2, A3}
- How many features subsets?
 - {A1}, {A2}, {A3}
 - {A1, A2}, {A1, A3}, {A2, A3}
 - {A1, A2, A3}

$$7 = 2^3 - 1$$

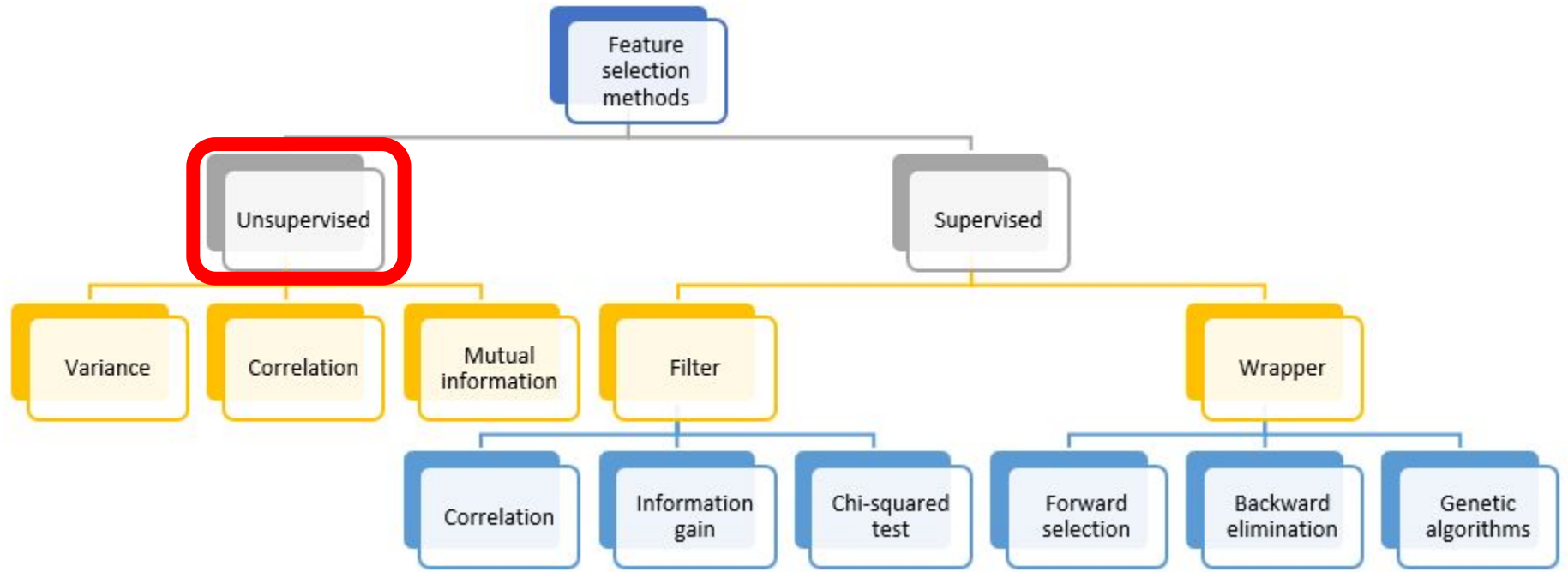
Why Feature selection is challenging?

- Number of features = n
 - {A1, A2, A3, ..., An}
- How many features subsets?
 - {A1}, {A2}, {A3}
 - {A1, A2}, {A1, A3}, {A2, A3}....
 - {A1, A2, A3}
 -
- For $n=100$

$$2^n - 1$$

1 267 650 600 228 229 401 496 703 205 376

Taxonomy of Feature Selection methods



Unsupervised Feature Selection

Unsupervised feature selection typically uses **statistical measures** to evaluate features:

- **Variance:**
 - Features with low variance are less likely to be informative and useful for prediction.
- **Correlation:**
 - Features that are highly correlated with each other are likely to contain redundant information.
- **Mutual information:**
 - Measures the amount of information that two features share.
 - Features with high mutual information are more likely to be predictive of each other.

Variance Filter

- Remove features with a high percentage of identical values across all objects.

	Feature 1	Feature 2	Feature 3	Feature 4
Object 1	4	8	0	16
Object 2	17	34	0	7
Object 3	4	8	0	3
Object 4	15	30	0	6
Object 5	8	16	0	4
Object 6	15	30	0	7

- Low variance indicates that constant features lack useful information.

Correlation Filter

- Remove highly correlated input numerical features.

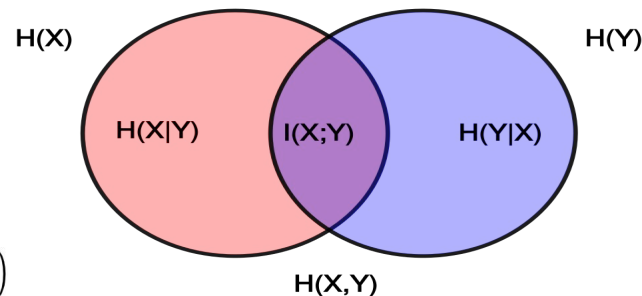
	Feature 1	Feature 2	Feature 3	Feature 4
Object 1	4	8	0	16
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Object 3	4	8	0	3
Object 4	15	30	0	6
Object 5	8	16	0	4
Object 6	15	30	0	7

- Correlated features indicates redundant information (Feature 2 = $2 \times$ Feature 1).
- Note:** Correlation can only detect linear relationships between features.

Mutual Information filter

- Measures the amount of information that two categorical features share using entropies.
- Features with high mutual information are more likely to be predictive of each other.
- Mutual information can detect any kind of relationship between two features.

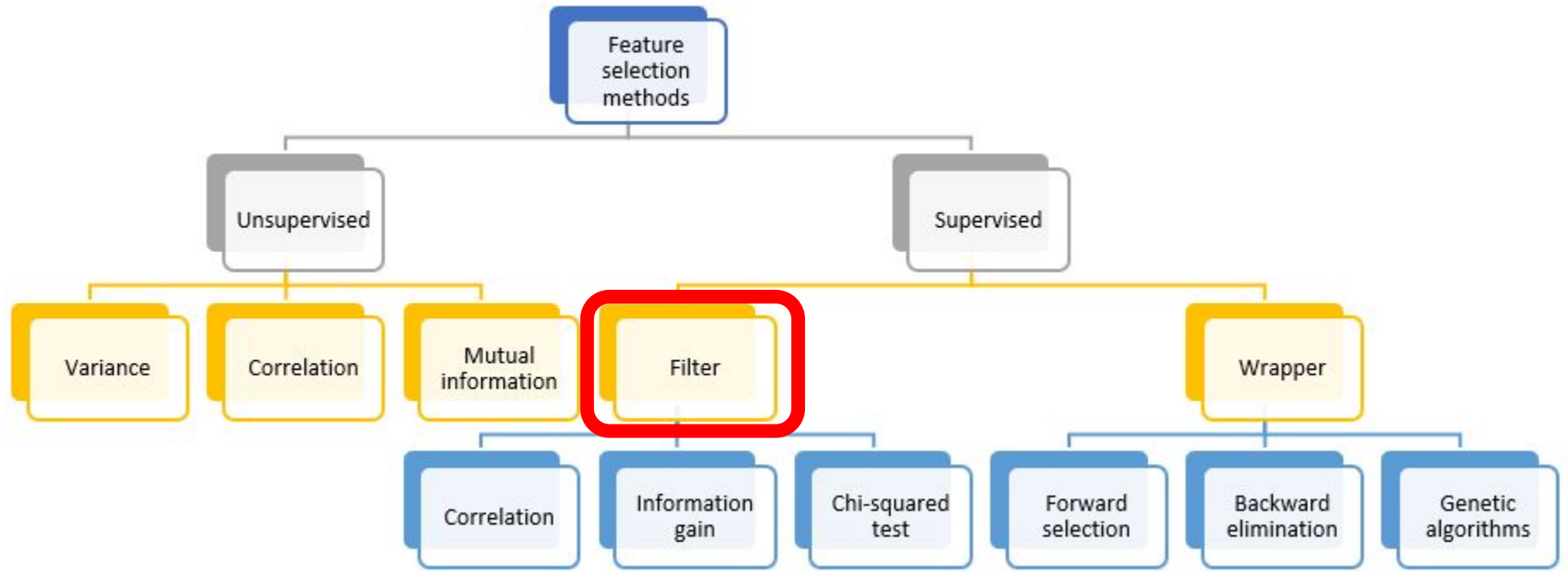
- ❖ $H(X)$, $H(Y)$: individual entropies
- ❖ $H(X, Y)$: joint entropies
- ❖ $I(X; Y)$: mutual information



$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) = H(X) + H(Y) - H(X, Y)$$

- The concept of entropy will be introduced in the upcoming slides.

Taxonomy of Feature Selection methods



Filter methods for FSS

Goal: evaluate features independently using the target variable, and select the most important ones based on statistical measures

Common statistical measures:

Correlation with the target variable, information gain, and chi-square test.

How to use filter methods for FSS

1. Choose a filter metric (correlation, information gain, or chi-square test).
2. Evaluate each feature using the chosen filter metric and rank them accordingly.
3. Select the features with the highest filter metric scores.

Filter methods: Correlation

- In filter methods, correlation measures how well a feature relates to the target variable.
- **Features highly correlated with the target variable:**
 - These features are often good candidates for prediction.
- **Features Lowly correlated with the target variable:**
 - These features may not contribute meaningfully to predictions.

Note: correlation does not consider the non-linear correlation with the target variable and the interaction between features.

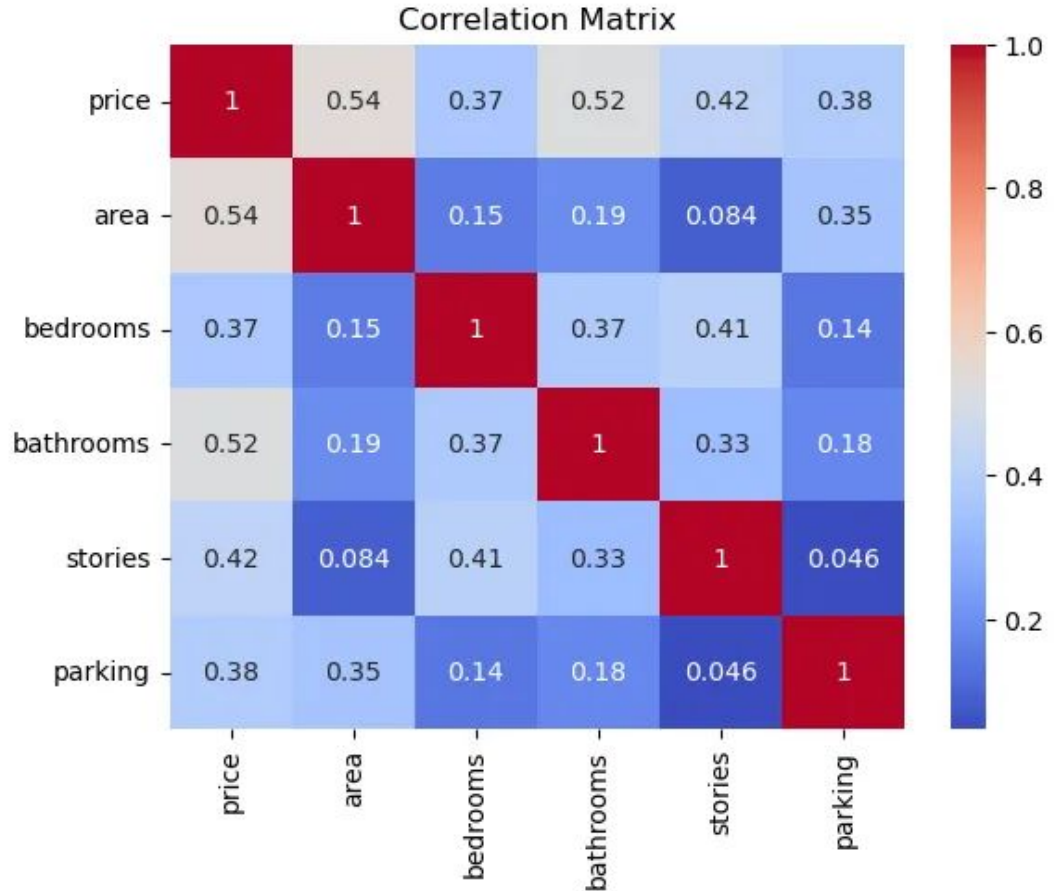
Correlation Heatmap

Helps in visualizing the correlation between attributes as well as the correlation with the target variable.

- Warm colors for positive.
- Cool colors for negative.
- Neutral for no correlation.

Example:

In this heatmap, suppose the price of the house is the target variable.



Filter methods: Information Gain

Goal:

Measure of how much information is gained about the target categorical variable when a dataset is split on a given categorical feature.

Principle:

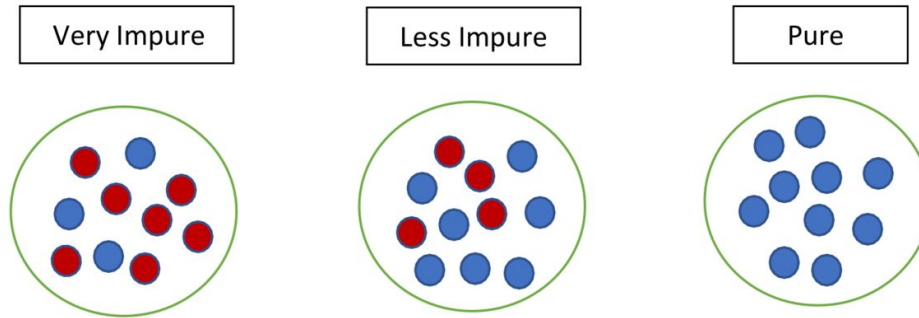
Calculate the difference between the **entropy** of the **target variable** before and after the split.

Note: Information Gain is commonly used in classification algorithms such as decision trees.

what is entropy?

What is entropy ?

- It measures the level of **uncertainty**, **randomness**, or **disorder** in a system or dataset.
- In the context of data, entropy quantifies the **impurity** of a set.



What is entropy ?

- It measures the level of **uncertainty**, **randomness**, or **disorder** in a system or dataset.
- In the context of data, entropy quantifies the **impurity** of a set.

$$H(S) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i)$$

- $S = \{x_1, x_2, \dots, x_n\}$, which is the set of elements
- $H(S)$: is the entropy of the whole set
- $p(x_i)$: is the probability of having the element x_i

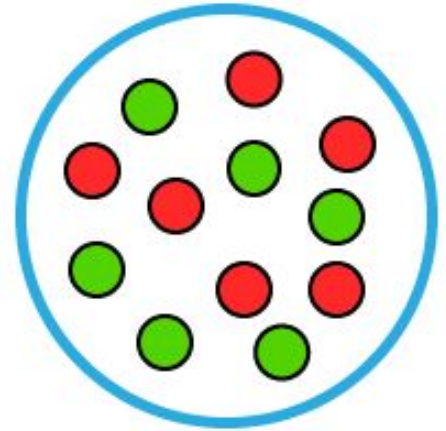
What is entropy ?

$$H(S) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i)$$

- $p(\text{red}) = 6/12 = 0.5$
- $p(\text{green}) = 6/12 = 0.5$
- $H(S) = -(0.5 \times \log(0.5) + 0.5 \times \log(0.5)) = -(-0.5 - 0.5)$

$$H(S) = 1$$

Very Impure



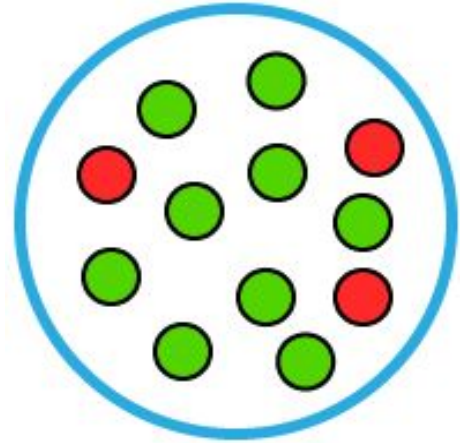
What is entropy ?

$$H(S) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i)$$

- $p(\text{red}) = 3/12 = 0.25$
- $p(\text{green}) = 9/12 = 0.75$
- $H(S) = -(0.25 \times \log(0.25) + 0.75 \times \log(0.75))$

$$H(S) = 0.811278$$

Less Impure



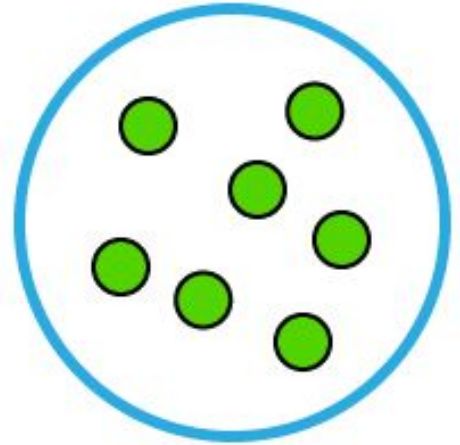
What is entropy ?

$$H(S) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i)$$

- $p(\text{red}) = 0$
- $p(\text{green}) = 1$
- $H(S) = -(0 + 1 \times \log(1))$

$$H(S) = 0$$

Minimum Impurity



Filter methods: Information Gain

Goal:

Measure of how much information is gained about the target categorical variable when a dataset is split on a given categorical feature.

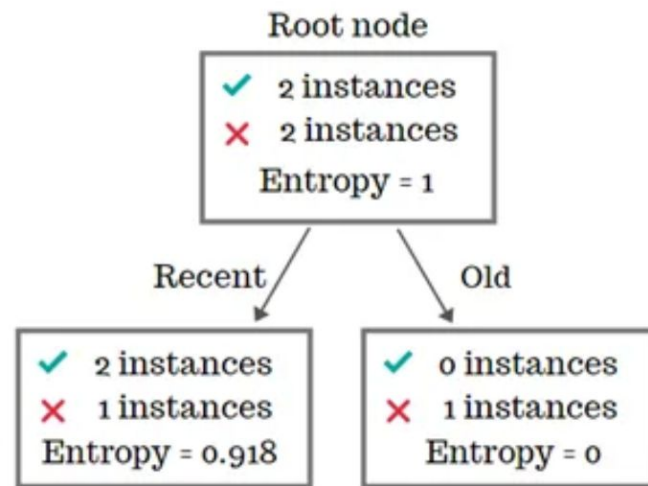
Principle:

Calculate the difference between the **entropy** of the **target variable** before and after the split.

$$IG(X) = H(S) - \sum_{i=1}^n p(x_i) H(S|x = x_i)$$

Example: Information Gain for Age

Age	Mileage	Road Tested	Buy
Recent	Low	Yes	Buy
Recent	High	Yes	Buy
Old	Low	No	Don't buy
Recent	High	No	Don't buy



- $$\begin{aligned} IG(\text{Age}) &= H(\text{Root node}) - (p(\text{Age} = \text{Recent}) \times H(\text{Recent})) - (p(\text{Age} = \text{Old}) \times H(\text{Old})) \\ &= 1 - ((\frac{3}{4}) \times 0.918) - ((\frac{1}{4}) \times 0) \end{aligned}$$

$$IG(\text{Age}) = 0.3115$$

Example: Information Gain for Road Tested

Age	Mileage	Road Tested	Buy	
Recent	Low	Yes	Buy	Pure set
Recent	High	Yes	Buy	
Old	Low	No	Don't buy	Pure set
Recent	High	No	Don't buy	

- $IG(\text{Road Tested}) = 1 - (0.5 \times 0) - (0.5 \times 0)$
 $= 1 - 0$

$IG(\text{Age}) = 1$ (Perfect predictor)

Filter methods: Chi-Square test

Goal:

Measure the association or dependency between a given **categorical feature** and the **target categorical variable**.

Principle:

Compare observed (**O**) and expected frequencies (**E**) to assess feature-target association.

Chi-Square

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Expected value for row i and column j

$$E_{ij} = \frac{(\text{Row Total of } i \times \text{Column Total of } j)}{\text{Grand Total}}$$

Example: Chi-Square test

Frequencies			
	Cat	Dog	
Men	207	282	489
Women	231	242	473
	438	524	962

Expected values			
	Cat	Dog	
Men	$\frac{489 \times 438}{962}$	$\frac{489 \times 524}{962}$	489
Women	$\frac{473 \times 438}{962}$	$\frac{473 \times 524}{962}$	473
	Cat	Dog	
Men	222,64	266,36	489
Women	215,36	257,64	473
	438	524	962

Dataset: information about people, one of the attribute is gender, we want to predict pets preference

Question: Is there a relationship between gender and preference for pets (cats or dogs)?

Assumption H0: The two columns are not related to each other.

Example: Chi-Square test

	χ^2		
	Cat	Dog	
Men	$\frac{(207-222,64)^2}{222,64}$	$\frac{(282-266,36)^2}{266,36}$	489
Women	$\frac{(231-215,36)^2}{215,36}$	$\frac{(242-257,64)^2}{257,64}$	473
	438	524	962

		X²		
		Cat	Dog	
Men	1,099	0,918	489	
Women	1,136	0,949	473	
	438	524	962	

$$\chi^2 = 1,099 + 0,918 + 1,136 + 0,949 = 4,102$$

- Define Degree of freedom: **df** = (number of rows-1)×(number of columns-1) = 1
- Define significance level α (probability threshold below which we reject **H0**, often $\alpha=0.05$)
- Look up the [Critical values of chi-square \(right tail\)](#), we find that the critical value is 3.84.
- $\chi^2 > 3.84$, so, we reject **H0**.

Filter methods for FSS

Advantages

- **Computational Efficiency**

No data mining model is trained $\Rightarrow$
Computational efficiency and scalability to
high dimensions and large datasets.

- **Algorithm Independence**

Independent of any data mining algorithm $\Rightarrow$
Unbiased toward any particular algorithm.

Disadvantages

- **Suboptimal Selection**

May not always identify the best feature
subset because it does not consider the
performance of the data mining algorithm.

- **Complex Relationships**

May not handle complex relationships and
interactions between features.

How to use filter methods for FSS

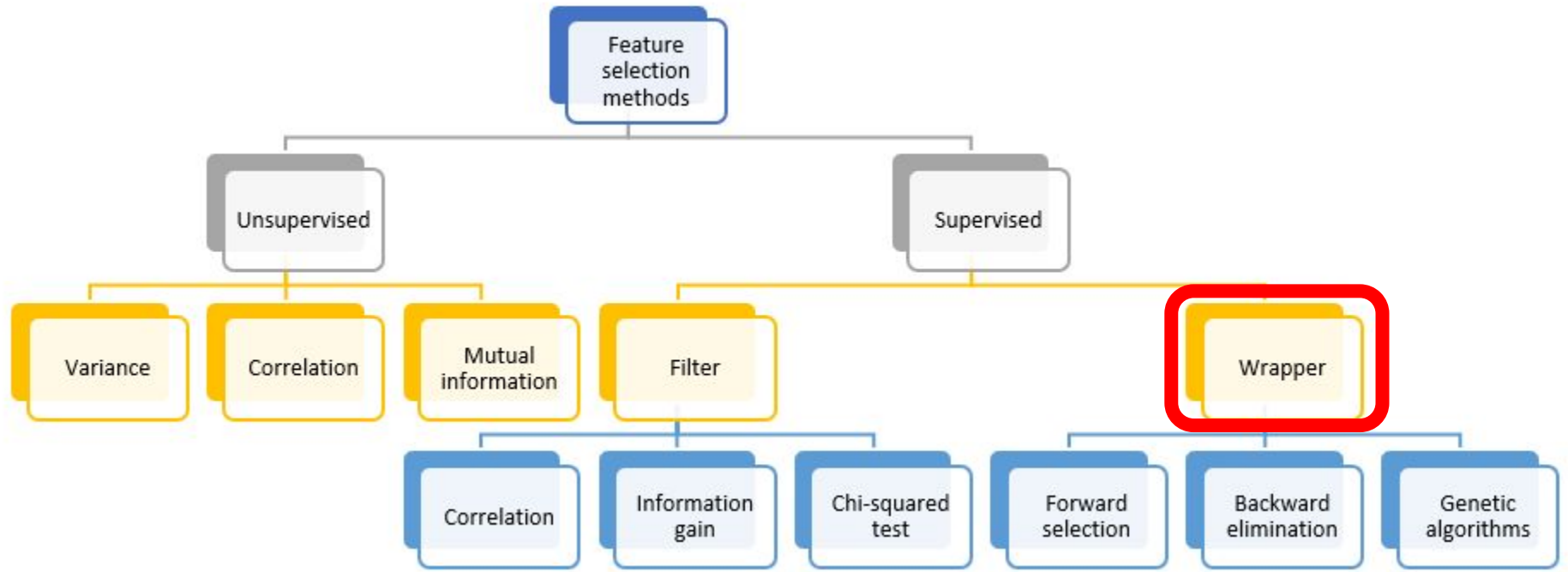
- **Filter Methods as an initial step**

- Filter methods serve as an excellent initial step in the feature selection process.
- Particularly valuable when dealing with large datasets.

- **Hybridization is key**

- It's advisable to combine filter methods with other feature selection techniques.
- Wrapper methods complement filter methods effectively.

Taxonomy of Feature Selection methods



Wrapper methods for FSS

Goal: evaluate features by using a data mining model as a black box.

Principle: the model is trained on different subsets of features, and the subset that produces the best performance is selected.

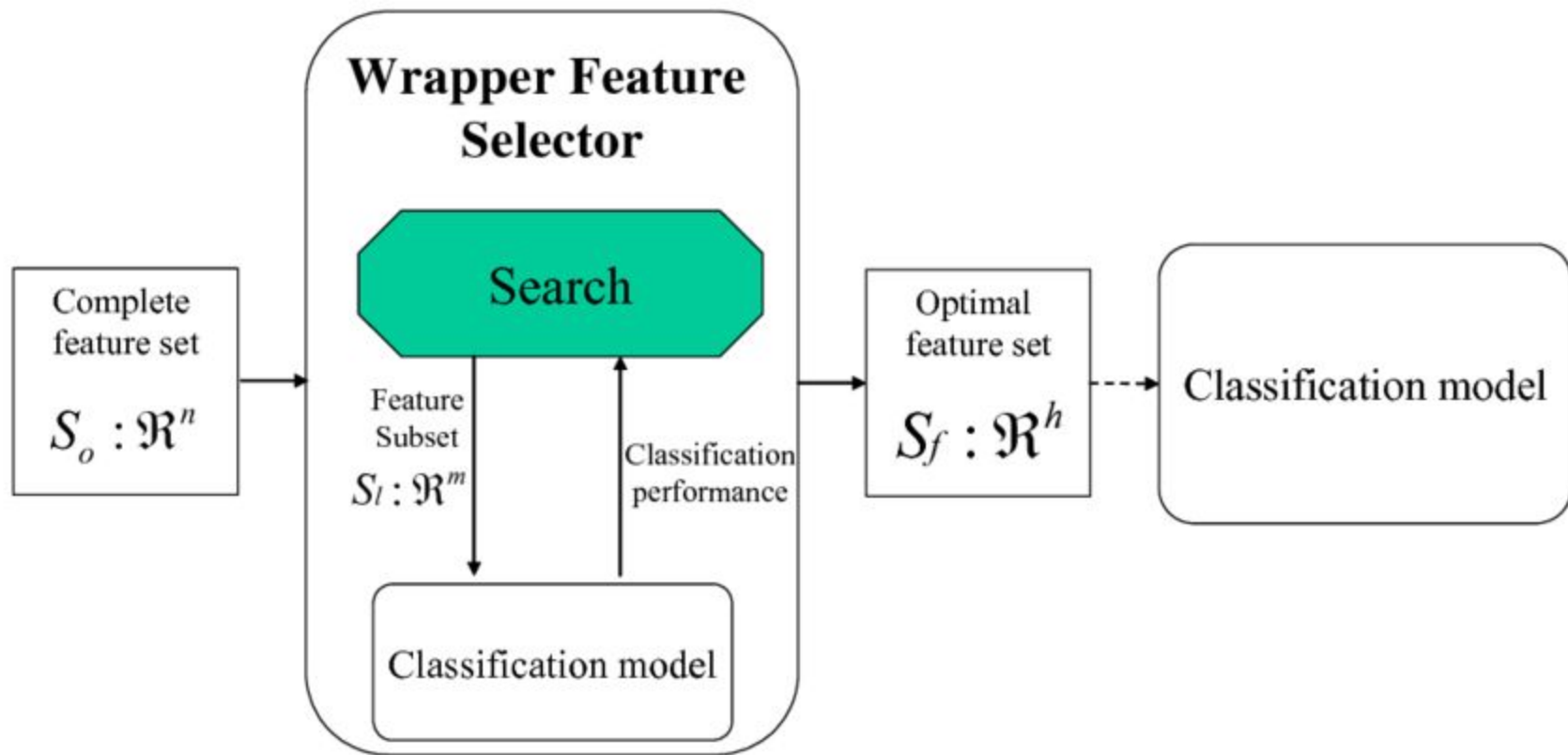
How to use filter methods for FSS

Feature Subset Generation: The search algorithm generates various feature subsets.

Model Evaluation: Each subset is input to the model and its performance is evaluated via model performance.

Selection Criterion: A selection criterion (e.g. accuracy) guides the choice of the best-performing feature subset

Iteration: The process is repeated with different subsets until an optimal set is found.



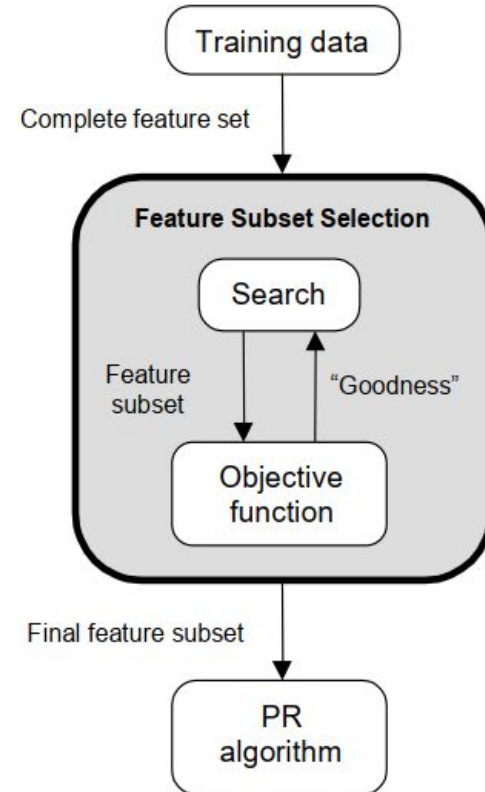
Search strategy and Objective function

- **Search Strategy:**

- Exhaustive evaluation of all subsets is impossible
- A search strategy navigates the vast feature subset space efficiently

- **Objective Function:**

- Evaluates candidate feature subsets
- Quantitative measure of the "goodness" or quality of a subset
- Feedback from the objective function to guide the search strategy



Search strategy and Objective function

- **Forward selection**

Add features step by step, choosing the one that improve the model the most at each iteration.

- **Backward elimination**

Remove features step by step, choosing the one that does not improve the model at each iteration.

- **Genetic Algorithm (GA)**

Optimization technique using natural selection to evolve feature subsets for better model performance.

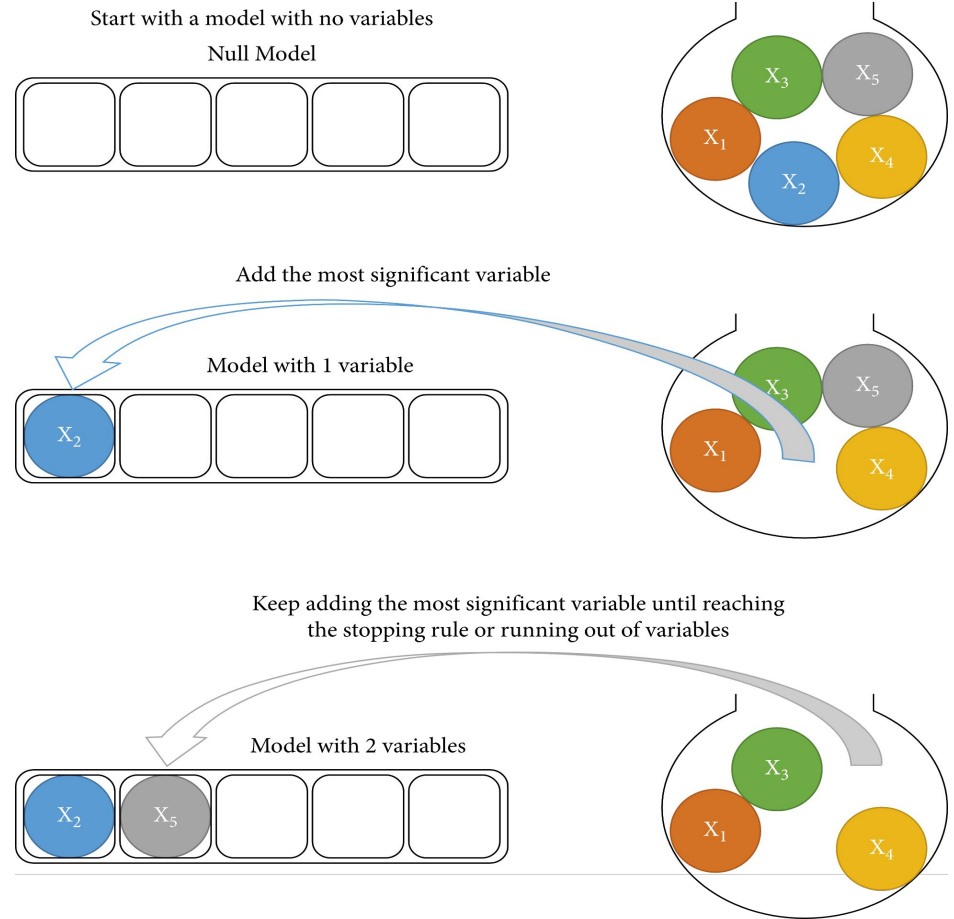
Forward Selection

1. Start with 0 features

2. Add the best feature

3. Stop adding features

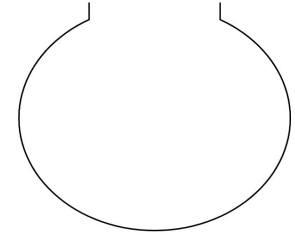
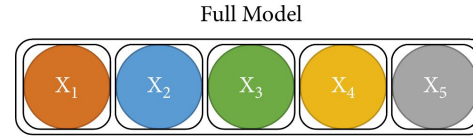
- a. Reach a given number of features.
- b. Performance threshold.
- c. Elbow methods.



Start with a model that contains all the variables

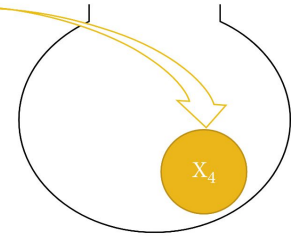
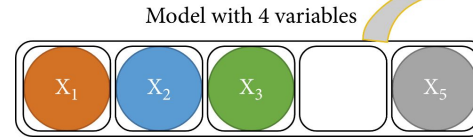
Backward elimination

1. Start with all feature



2. Remove the worst feature

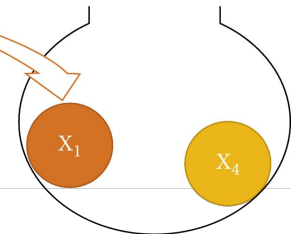
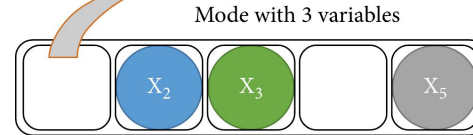
Remove the least significant variable



3. Stop removing features

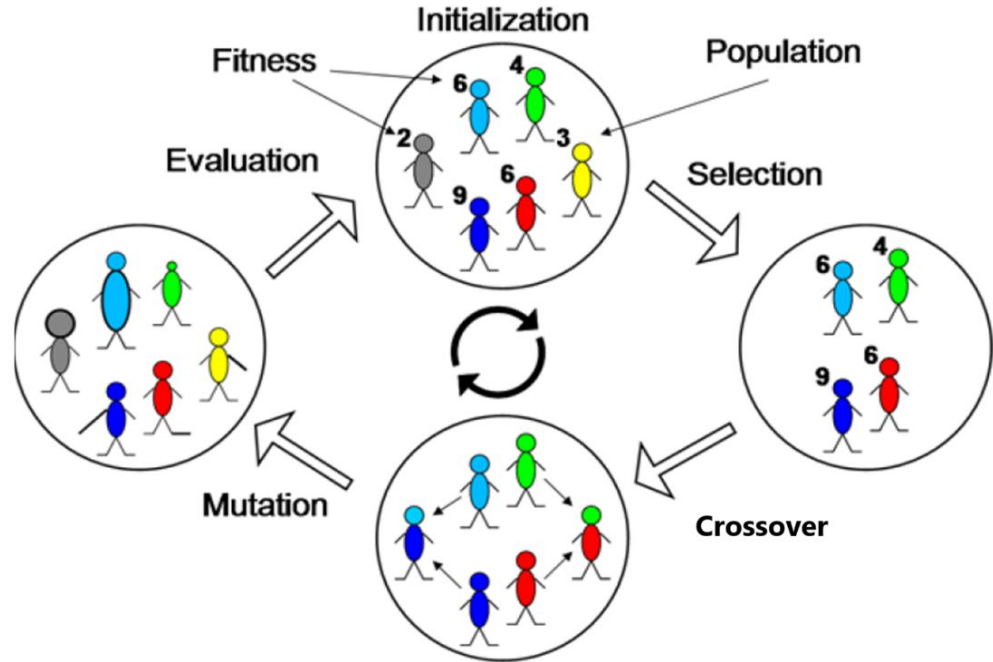
- Reach a given number of features.
- Performance threshold.
- Elbow methods.

Keep removing the least significant variable until reaching the stopping rule or running out of variables



Genetic Algorithm (GA) for Feature Selection

What is a genetic algorithm?



Genetic Algorithm (GA) for Feature Selection

- **Crossover**

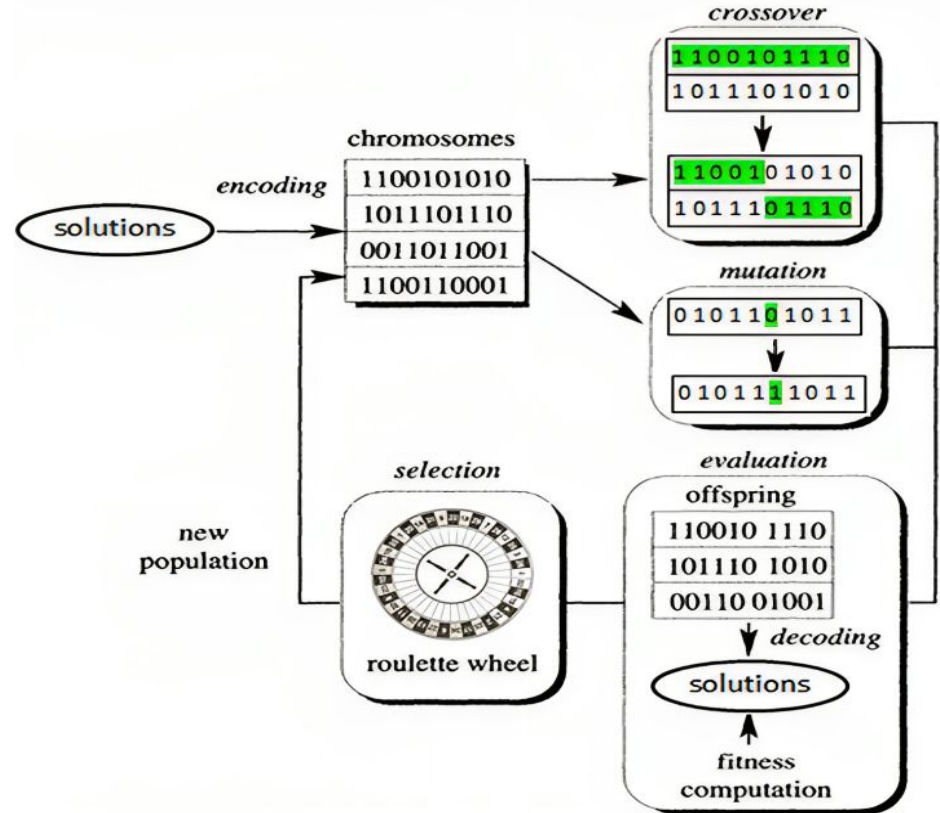
- Combine two solutions
- Produce good children

- **Mutation**

- Random change
- Exploration

- **Selection**

- Select the best
- Survival of fittest



N.B. In feature selection and GA, we can encode the selected feature using 1's and the discarded one by 0's

Wrapper methods for FSS

Advantages

- **Optimal Subset Selection**

Can detect the ideal feature subset for a given data mining algorithm.

- **Managing Complex Relationships**

Effective in handling relationships between features by evaluating a subset, not a feature.

Disadvantages

- **Computational Intensity**

Tend to be computationally demanding, particularly with large datasets.

- **Algorithm Bias**

May exhibit bias towards the data mining algorithm used for feature evaluation.

Comparison of Feature Selection Methods

Method	Advantages	Disadvantages
Unsupervised	- Simple - Fast - No need for labels	- May miss complex relationships - May not directly improve model performance - May lead to irrelevant features
Filter methods	- Fast - Algorithm independent - Easily interpretable	- May miss complex relationships - Suboptimal for specific algorithms - Doesn't account for interaction between features
Wrapper methods	- Finds optimal subset for the algorithm - Handles complex relationships	- Slow - Algorithm bias - Computationally expensive for large feature sets
Forward selection	- Adds features incrementally	- Suboptimal for specific algorithms - May miss Feature Interactions
Backward elimination	- Removes features incrementally	- Can be slow for high-dimensional data - Risk of overfitting with many features
Genetic algorithms	- Thorough search - Can handle complex interactions - Can explore a large search space	- Computationally intensive - Risk of overfitting - Difficult to tune parameters

Comparison of Feature Selection Methods

How to choose a Feature Selection Method?

- **Size of dataset**
 - Filter methods work well for large datasets
 - Wrapper methods work well for small datasets
- **Computational budget**
 - Filter methods are fast
 - Wrapper methods are slow
- **Need for interpretability**
 - Filter methods allow inspection of feature importance
 - Wrapper methods act as black box
- **Domain knowledge**
 - Unsupervised methods do not use labels
 - Supervised methods use domain labels
- **Algorithm fit**
 - Wrapper methods tailor to specific DM algorithm
 - Filter methods are independent of any algorithm
- **Feature interactions**
 - Wrapper methods handle feature interactions
 - Filter methods assess features independently

Summary

The goal of FSS is to select an optimal subset of features

- **Main approaches:**
 - Unsupervised: Fast evaluation of features using statistical metrics
 - Supervised Filter: Evaluate features independently using statistical metrics along with the target
 - Supervised Wrapper: Use model performance to search feature subsets
- **Considerations:**
 - Dataset size and dimensionality
 - Computational budget and time constraints
 - Need for interpretability vs performance
 - Domain knowledge of features
 - The chosen data mining algorithm
 - Relationships between features
- No one-size-fits-all method, choose a method based on goals, data, and constraints
- Hybrid approaches combine strengths of different techniques